

Order matters on tracks

Production detailed routers process nets in an order

The idea

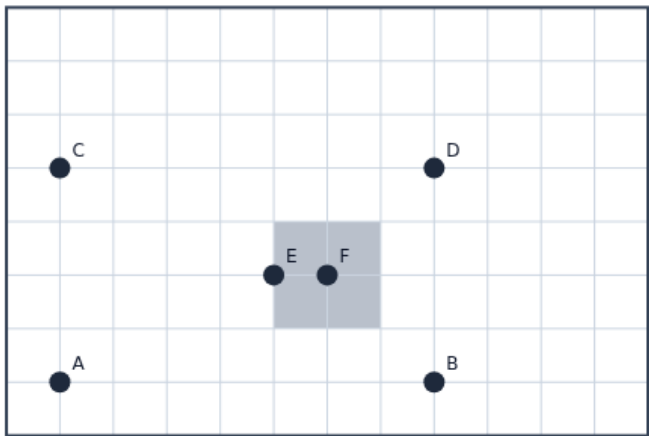
- Initialize empty usage
- Increment usage on every track key in the route
- Report final usage and track_overflow summary

Net order

SEQUENTIAL DETAILED

Net order

Step 1 / 5 · order · module04-01-sequential-detailed



Explanation

Route nets 0..5 in order; later nets see earlier usage.

- Six nets

NETS array

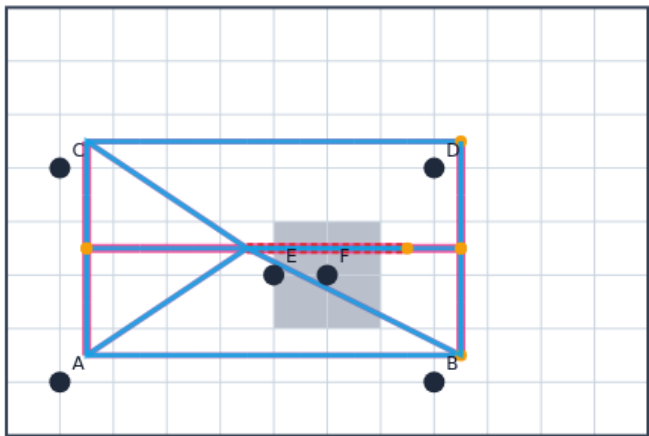
Net order

Layered pass

SEQUENTIAL DETAILED

Layered pass

Step 2 / 5 · layer-all · module04-01-sequential-detailed



Explanation

Layered L all nets documents baseline overflow.

- Pattern route

total≈3

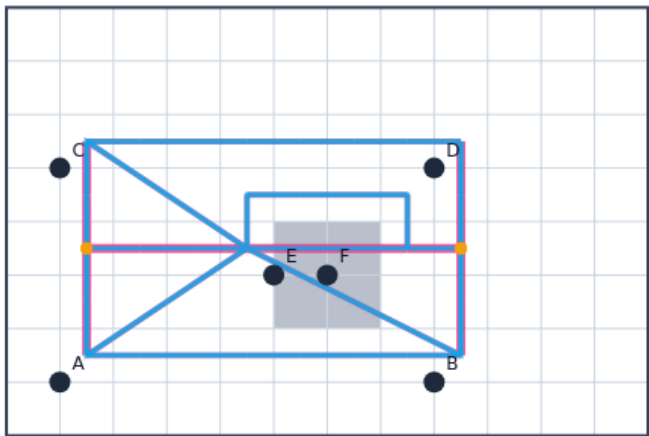
Layered pass

Lee pass

SEQUENTIAL DETAILED

Lee pass

Step 3 / 5 · lee-all · module04-01-sequential-detailed



Explanation

Sequential Lee may redistribute usage.

- Maze mode

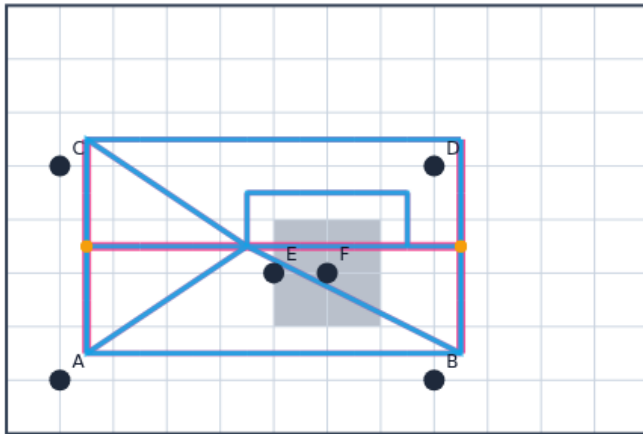
Lee≈0

Clear overflow

SEQUENTIAL DETAILED

Clear overflow

Step 4 / 5 · clear · module04-01-sequential-detailed



Explanation

Goal: total overflow 0 after A* or moves.

- Challenge lab

HPWL optional

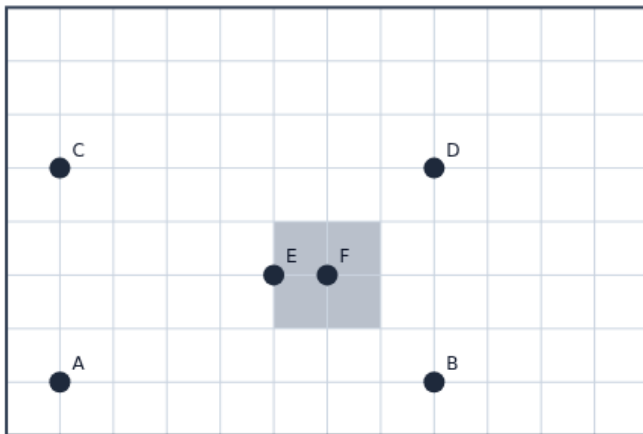
Clear overflow

Handoff

SEQUENTIAL DETAILED

Handoff

Step 5 / 5 · done · module04-01-sequential-detailed



Explanation

Detailed routing feeds signoff DRC and extraction.

- Tracks + vias

learn_drc next

Handoff

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Sequential detailed routing

INTERACTIVE TOOL

Sequential detailed routing

Route all six nets in order; clear overflow with layered, Lee, or A*.

Your job: route all 6 nets sequentially. Try layered, Lee, or A* until overflow clears.

Reset workspace

Challenges0 / 8 cleared

Pick oneIntro Six nets

Pick a challenge and click **Start**. Move cells, route, then **Check**.

StartShow hintCheckNext challengeStop checkingIdle

A B C D E F←→↑↓Route LeeRoute layeredRoute A*

Rip-up detailed

Graph

Reference / metrics

selected: A @ grid (1,1)
routes: (none – click Route)
overflow total: 0.00 · max: 0.00 · count: 0
track capacity: 2 · mode: astar
HPWL: 34.0

Browser lab starter

Implement track

- Implement ``sequential_detailed`` and ``route_from_data``
- Print usage and overflow after routing all six nets
- Compare astar mode with `l_hv` on total overflow

Pitfalls

- Parallel deposit without order, hides rip-up motivation
- Ignoring multi-pin net in order list
- Resetting usage between nets accidentally

Your turn

- Complete sequential detailed routing
- Offline compare and wrap come next